

● MEDIUM

● **ADVANCED MATH**

Advanced Math

10 questions · ~15 min

14

Q1

ADVANCED MATH

$$T = 0.01(P - 40,000)$$

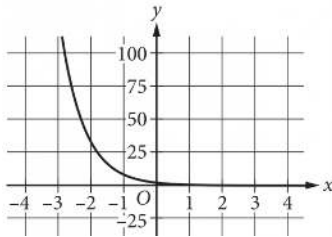
In a city, the property tax T , in dollars, is calculated using the formula above, where P is the value of the property, in dollars. Which of the following expresses the value of the property in terms of the property tax?

- A. $P = 100T - 400$
- B. $P = 100T + 400$
- C. $P = 100T - 40,000$
- D. $P = 100T + 40,000$

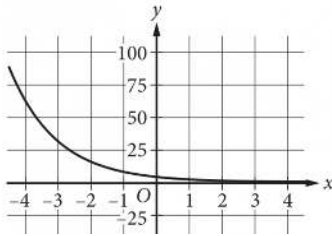
$$y = 4(2^x)$$

Which of the following is the graph in the xy-plane of the given equation?

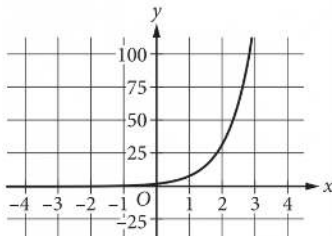
A.



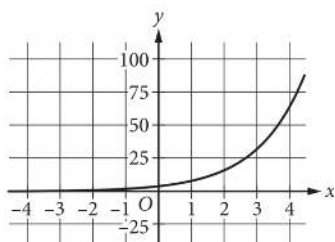
B.



C.



D.



Q3

ADVANCED MATH

$$-4x^2 - 7x = -36$$

What is the positive solution to the given equation?

- A. $\frac{7}{4}$
- B. $\frac{9}{4}$
- C. 4
- D. 7

Q4

ADVANCED MATH

The function f is defined by $fx = 2700.1^x$. What is the value of $f0$?

- A. 0
- B. 1
- C. 27
- D. 270

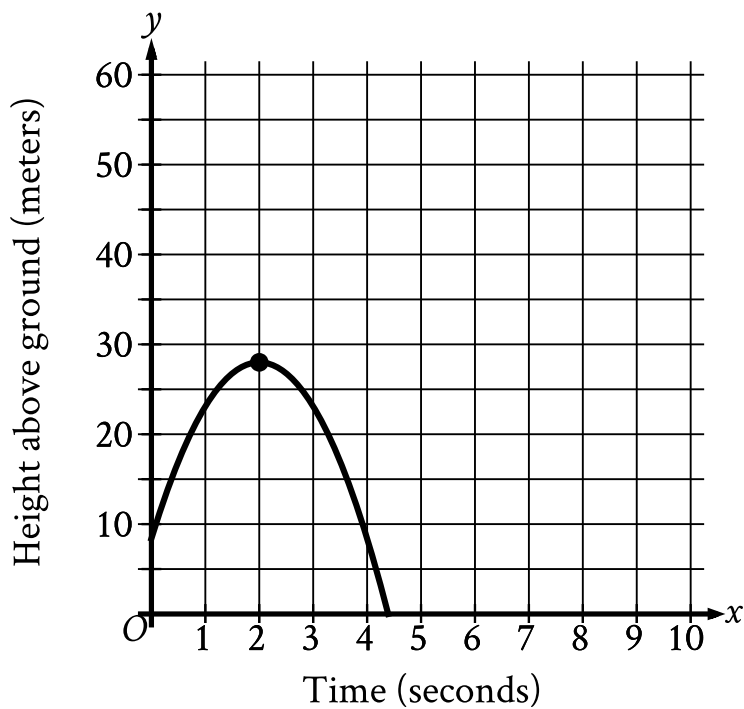
Q5

ADVANCED MATH

$$3xx - 4x + 5 = 0$$

What is one of the solutions to the given equation?

- A. -4
- B. 0
- C. 3
- D. 5



- The parabola opens downward.
- The vertex is at point (2 comma 28).
- The parabola passes through the following points:
 - (0 comma 8.4)
 - (2 comma 28)
 - (4 comma 8.4)

An object was launched upward from a platform. The graph shown models the height above ground, y , in meters, of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

- A. From $x = 0$ to $x = 2$
- B. From $x = 0$ to $x = 4$

C. From $x = 2$ to $x = 3$

D. From $x = 3$ to $x = 4$

Q7

GRID-IN

ADVANCED MATH

$$z^2 + 10z - 24 = 0$$

What is one of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

ADVANCED MATH

The number of bacteria in a liquid medium doubles every day. There are 44,000 bacteria in the liquid medium at the start of an observation. Which represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

A. $y = \frac{1}{2}44,000^t$

B. $y = 244,000^t$

C. $y = 44,000\frac{1}{2}^t$

D. $y = 44,0002^t$

Q9

ADVANCED MATH

$$y = x - 2x + 4$$

$$y = 6x - 12$$

Which ordered pair x, y is the solution to the given system of equations?

A. 0, 2

B. -4, 2

C. 2, 0

D. 2, -4

The function f is defined by $fx = 4x^{-1}$. What is the value of $f21$?

A. -84

B. $\frac{1}{84}$

C. $\frac{4}{21}$

D. $\frac{21}{4}$